Instructions Buy the Phrase; the Phrase Costs Preference: an Epistemic Audit of LLM Aggregation Pipelines at Auditable Scale

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Abstract

Mixture-of-Agents (MoA) pipelines — proposer models whose outputs an aggregator synthesizes — achieve state-of-the-art results on preference-judged benchmarks, but every published gain is measured by an LLM preference judge, and nothing in that literature measures what such gains are made of. We audit the MoA architecture at auditable scale (7–20B local models) with pre-registered cells, validated instruments, and causal designs, and report three linked findings. (1) A phrase-swap experiment shows that an epistemic instruction ("label estimates with phrase X") produces *only* its named phrase: the dictated form appears in two thirds of outputs regardless of which synonym is named, while the construct rate outside the named phrase is statistically unchanged (CIs [−0.079, +0.111], [−0.103, +0.087]). (2) A preference judge, order-debiased, shows a real aggregation lift — the aggregated pipeline beats the same writer answering directly in 0.818 [0.718, 0.919] of decisive pairs, surviving length matching — yet *penalizes* the instructed arms: the phrase-bearing arms lose to their own control in both counterbalanced forms (0.261 [0.140, 0.388]; 0.317 [0.152, 0.480]). Because the arms differ only in the phrase, the penalty is causally attributable to it. Together these results identify a measured tension: instruction tuning of epistemic disposition installs surface forms, and preference optimization removes exactly those forms. (3) The preference instrument itself is majority reading-order at this scale: raw first-position preference is 0.85, order-debiasing converts two thirds of pairs to ties, and a second local judge produced 95% ties. We further report mechanism results that bound the MoA narrative — the aggregator selects sources rather than recomputing, ignores cross-proposer corroboration, and transports roughly 15% of upstream epistemic qualification — and release the measurement kit, the dictation registry, and all pre-registrations.

1 Introduction

Mixture-of-Agents [1] reports that a stack of open-source proposers plus an aggregator surpasses GPT-4o on AlpacaEval 2.0 (65.1% vs. 57.5% length-controlled win rate), and frames the result as "collaborativeness": models produce better responses when shown other models' outputs, even lower-quality ones. Every number in that paper — and in the surrounding multi-agent literature — is produced by an LLM preference judge. No published MoA evaluation measures a verifiable outcome, tests error propagation, validates its judge on the deployed task, or separates behavioral change from surface compliance.

This paper supplies that audit. It is the product of a falsification-first research program that ran forty-three pre-registered experiment cells on a MoA-architecture pipeline small enough to audit completely: 7–20B open models on consumer hardware, with every registration, deviation, and verdict recorded in a public ledger before and after each run. The architecture is a two-layer MoA with three proposers and one aggregator. Earlier program cells ran the MoA Aggregate-and-Synthesize prompt *verbatim*; the causal cells of §5–§6 deliberately use a minimal aggregator prompt instead, so that instruction content is the manipulated variable rather than a background condition.

Three questions organize the audit:

1. **What do epistemic instructions to an aggregator actually do?** (§5) A counter-balanced phrase-swap cell shows the answer is: install the named phrase, and nothing measurable beyond it.

2. **Does the MoA preference lift replicate at auditable scale, and what is it made of?** (§6) It replicates — 0.818 of decisive pairs, surviving a length control — and its composition resists our full surface feature set. The same judge penalizes instructed epistemic marking, causally.

3. **Do the mechanisms the MoA narrative assumes exist?** (§7) Largely no: the aggregator prefers uncorrupted sources but does not recompute; corroboration among proposers does not increase transport; two same-base domain fine-tunes do not share a reasoning signature.

The audit's instruments are themselves audited (§4): fourteen recorded instrument-validity findings drive the methodology, including the discovery — made mechanically by a dictation registry — that an earlier judge prompt in our own program enumerated the same phrases the system prompts dictated to writers, entangling instrument with scaffold at the instrument layer.

Contributions. (1) The first causal separation of compliance from behavior in epistemic instruction-following (the phrase-swap design), with a two-link causal chain showing preference optimization and epistemic disposition are in tension. (2) A replication of the MoA preference lift at 20B scale under an order-debiased protocol, with a decomposition showing it is not length, domain-framework density, or hedging register. (3) A measured characterization of local pairwise preference judging as majority reading-order. (4) Mechanism results bounding the collaborativeness narrative. (5) An open measurement kit (`gst`), a dictation registry with a zero measured over-attribution rate, and the complete pre-registration ledger.

2 Related work

Aggregation pipelines. MoA [1] is the direct object of this audit. [2] showed that aggregating repeated samples of the single best model ("Self-MoA") outperforms mixed MoA by 6.6% LC on AlpacaEval — converging, from the benchmark side, with our finding that mixing is not automatically diverse (§7). Ensemble alternatives (rankers, routers, fusers) are surveyed in [1]; our LLM-ranker-adjacent result is the source-selector finding. [6] report that a single strong-prompted agent can match multi-agent discussion, consistent with our council nulls.

Judge pathologies. Position bias and verbosity bias in LLM judges are documented by [3]; length correlations in preference optimization by [4]; sycophancy under preference pressure by

[5]. Our contribution is quantitative and causal at the pipeline level: an 85% raw first-position rate for a 20B judge, and a counterbalanced demonstration that specific dictated epistemic phrases lose preference.

Instruction following. Our phrase-swap design is, to our knowledge, the first to test an epistemic instruction by swapping its named surface form under otherwise byte-identical prompts, isolating form-independent behavioral residue from compliance.

3 The pipeline and the ledger

The audited pipeline is a two-layer MoA: three proposer seats (prompted or domain-fine-tuned 7–8B models, or role-prompted samples of the writer) whose outputs are concatenated for a 20B mixture-of-experts writer (`gpt-oss:20b`) under either the MoA Aggregate-and-Synthesize prompt or a minimal "lead analyst" prompt. The case battery is eighteen strategic advisory scenarios spanning healthcare, legal, and finance content.

Every cell is registered before it runs: hypotheses, falsification conditions, attainability computations (the fraction of runs in which the outcome variable can fire, computed from existing data, for every registered prediction), and consequences. Verdicts land in a status ledger that distinguishes LIVE, PROVISIONAL, WITHDRAWN, and BLOCKED claims; three program-level audits re-classified earlier results, and nulls are cited only at the strength their audit class permits (informative, weak, uninformative, or degenerate). All numbers below are LIVE ledger entries unless marked otherwise.

4 Instruments, audited

Fourteen recorded instrument-validity findings shape the methodology. Four matter most here.

Entanglement (finding 8). If a prompt dictates the strings an instrument detects, measurement M confounds compliance C with behavior B: $M = C + B$, and a perfect detector still counts an echoed phrase, because the difference between C and B is causal, not textual. A pattern-level decomposition (PD-13) first exposed this: an instruction moved its dictated phrasings from 0/30 to 28/30 while non-dictated phrasings of the same construct moved 1/30 to 4/30 (difference-in-differences +0.83 [+0.67, +1.00]).

The dictation registry. We extracted every construct-bearing phrase any system, generation, or judge prompt places in a model's mouth (125 entries from 35 prompt symbols; AST parsing and quote scanning, no semantic classifier), froze it under a hash, and gated all subsequent prompts against it. On its first run the registry found that an earlier pairwise judge prompt in our own program enumerated the same seven phrases the training-pair generators dictated to writers — instrument entanglement at the instrument layer (finding 12). Measured over-attribution is zero: across 2,907 clean-provenance spans (438,797 characters), no registry phrase appears without dictation upstream (finding 13). Registry surface is therefore diagnostic of the scaffold, which is what licenses literal-match partitioning.

The validated judge. Construct presence is scored by a two-judge, batched sentence-level protocol whose definitions name no phrase and instruct judges not to reward particular wording; it passes the registry gate, and agreement on the deployed corpus is 0.910 (14,088/15,485 decisions). Document-level judging of the same corpus scored 0.622 and was rejected (finding 9).

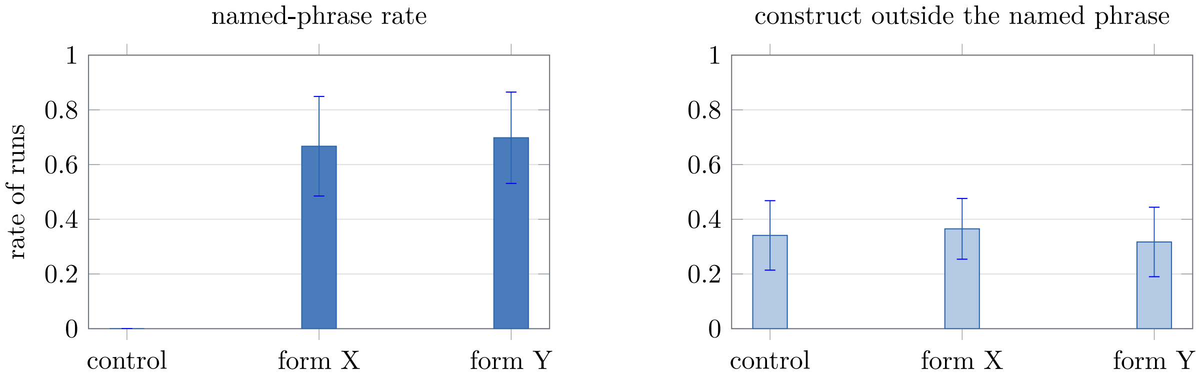

Figure 1: The phrase-swap cell (378 runs, cluster-bootstrap 95% CIs over 18 cases). **Left**: each instructed arm produces its own named phrase in about two thirds of runs; control never produces either, and cross-form leakage is 0/252. **Right**: the validated-judge construct rate *outside* the named phrase is statistically indistinguishable across all three arms. The instruction's entire measurable effect is the phrase it names.

Preference judging is majority reading-order (finding 14). In the preference cell (§6), the 20B judge's raw first-position preference was 0.85; judging every pair in both orders and requiring agreement converted 65% of pairs to ties; a 7B second judge produced 95% ties. Order-debiased protocols keep preference verdicts honest at the cost of roughly two thirds of the sample.

5 The phrase-swap cell: instructions buy the phrase

Design. Three arms, 18 cases, 7 repeats (378 runs). The control arm is the minimal writer prompt. The two treated arms append a byte-identical clause differing in exactly three words: *"When you state a number that is an estimate rather than an established fact, label it using the phrase 'modeled at' "* (form X) or *"…'taken to be' "* (form Y). Form X is the historically dictated house form (11 registry entries across 9 prompt symbols); form Y was chosen the day before the run, appears zero times in 2M characters of archive, and matches form X's spontaneous rate exactly (0/249 clean-provenance runs for both). A harness gate refuses to run unless each arm contains exactly its own form and no other registry phrase.

Compliance is form-tracking and the house form has no privilege. Form X appears in 0.667 of its arm's outputs (bootstrap CI vs. control [+0.468, +0.857]); form Y in 0.698 ([+0.524, +0.865]); cross-form leakage is 0/252 — no treated run ever produced the other arm's form, and no control run produced either (Figure 1, left). A phrase drilled through months of system prompts and a day-old synonym perform identically: whatever the instruction engages, it is not a learned disposition toward the trained-on words.

Nothing survives the swap. With each arm's own named phrase subtracted, the validated-judge construct rate is statistically unchanged: form X 0.341 → 0.365 ([−0.079, +0.111]); form Y 0.341 → 0.317 ([−0.103, +0.087]) — opposite nominal directions, both spanning zero (Figure 1, right). Emission is flat across arms (43.7–46.6 sentences), so the null is not bought with silence. The registered minimum detectable effect is +0.224 at the realized cluster count; the result licenses "no form-independent effect ≥ 0.22," not "zero."

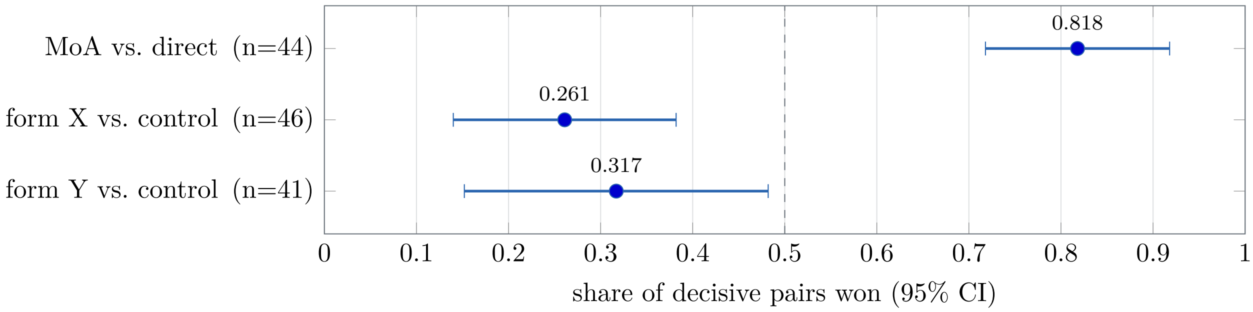

Figure 2: The preference cell (order-debiased; a pair is decisive only when the same side wins both orderings; cluster-bootstrap CIs over cases). The aggregated pipeline beats the matched-length direct baseline decisively; both phrase-instructed arms *lose* to their own control, in counterbalanced forms whose only difference from control is the phrase. The dashed line is indifference.

The instruction's entire measurable effect is the phrase it names.

6 The preference cell: the lift is real, and the phrase costs preference

Design. Three pairwise comparisons on the same 18 cases (126 pairs each; both orderings; decisive only when the same side wins both; primary judge `gpt-oss:20b`, which also wrote every output on both sides of every pair, making self-preference symmetric by construction). C1 pairs the phrase-swap control arm (the 2-layer MoA) against a generated direct arm: the same writer, same cases, same sampling, with the writer prompt minus only the sentence referencing contributions. C2 and C3 pair each instructed arm against control — and because §5 established those arms differ only in the phrase, any preference gap is causally the phrase's.

The lift replicates at 20B scale. The aggregated side wins 0.818 [0.718, 0.919] of 44 decisive pairs. Arm mean lengths differ by 1.4% (7,757 vs. 7,649 chars), and the share among near-length-matched pairs is 0.815, so the lift is not length (Figure 2). It also does not track domain-framework density (the winner has more frameworks per kilocharacter in only 0.409 of decisive pairs). Its composition is *unidentified* by our feature set — reported as such.

The phrase arms lose, in both forms. The instructed arms' shares against their own control are 0.261 [0.140, 0.388] (form X) and 0.317 [0.152, 0.480] (form Y): both confidence intervals lie entirely below 0.5, in the same direction, across counterbalanced forms, and the penalty survives length matching. Our registered prediction was the opposite (that judges reward the compliance register); the result is a falsification-by-reversal, and is stronger than the prediction: **dictated epistemic marking costs preference points**.

The tension, stated. Chaining §5 and §6, both links causal: an epistemic instruction installs only a surface form; a preference judge penalizes that form. A pipeline tuned toward judge preference — RLHF-style or best-of-n against a preference model — will therefore shed exactly the epistemic marking that instruction paradigms install, while leaving the underlying disposition (unchanged by the instruction anyway) untouched. Preference optimization and epistemic disposition are not merely unaligned; at this scale they are measurably opposed at the surface where instructions act.

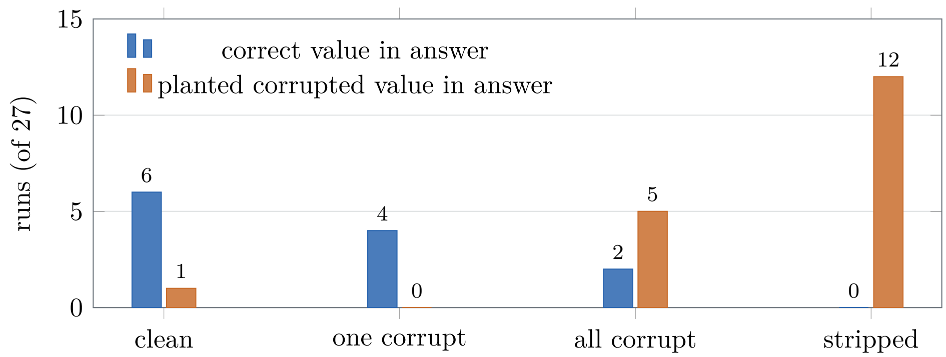

Figure 3: The source-selector gradient (planted arithmetic errors; exact match on known values). As clean sources are removed — one proposer corrupted, all corrupted, working stripped — adoption of the planted wrong value rises monotonically $1 \to 0 \to 5 \to 12$ while correct values vanish. The aggregator prefers uncorrupted sources when they exist and has no independent hold on the content when they do not.

7 Mechanisms the MoA narrative assumes

The aggregator selects sources; it does not recompute. With arithmetic claims planted in proposer texts: when one proposer is corrupted and a clean one remains, the corrupted value reaches the answer 0/27 times; when every proposer is corrupted, 5/27; when the working is also stripped, 12/27. The monotone gradient $1 \to 0 \to 5 \to 12$ across arms (Figure 3) identifies preference for uncorrupted sources with no independent recomputation. MoA's own evidence that the aggregator's output overlaps most with the best proposals [1] is the same phenomenon viewed from the benign side; the planted-error view adds its failure boundary, and shows the "collaborativeness even from worse inputs" claim fails at the fact level: worse inputs made the writer factually worse.

Corroboration is ignored. Claims advanced by $k \geq 2$ independent proposers transport to the final answer no more often than single-proposer claims (0.487 vs. 0.524; difference CI [−0.153, +0.080], excluding gains above +0.08) — the one reliability signal the architecture uniquely provides is unused.

Transport is weak; invention is real. On de-scaffolded arms under the validated instrument, the writer transports upstream epistemic qualification with slope $w = 0.150$ [0.058, 0.252] (one writer; second writer's CI includes zero at n=20), and invents qualification at a 0.333 [0.138, 0.609] rate on inputs confirmed to contain none.

Mixing is not automatically diverse. Two medical fine-tunes of the *same* base model do not share a domain signature: one shifts in-domain framework density +0.232 [+0.072, +0.412] over the base, the other +0.109 [−0.056, +0.300], and a second lineage shows −0.041 [−0.099, +0.025] — per-model, not per-domain-training. A post-hoc descriptive check (not a registered result) further found two same-domain seats no more distant from each other than one seat is from itself across samples; registered or not, the published Self-MoA result [2] independently reaches the same conclusion at benchmark scale.

Accuracy. On a 60-item self-verifying battery the aggregated pipeline and the bare writer are statistically indistinguishable, but the comparison sits at a 0.92–0.97 ceiling with three discordant items; per our audit discipline this is held as absence of evidence, not evidence of absence.

8 Discussion

What the lift is, and is not. The audit does not debunk MoA's headline: aggregation genuinely wins preference at our scale, under a debiased protocol, against a matched-length direct baseline. What it removes is the assumed mechanism story. The lift is not built from the epistemic behaviors the aggregate prompt requests (those requests install phrases, and the judge penalizes them), not from length, and not from domain-framework density; and the aggregator beneath it selects rather than verifies. The residual — plausibly organization, coverage, or specificity harvested from proposers — is the right target for the next instrument, and we decline to name it without one.

Implication for preference-trained pipelines. The measured tension predicts that preference optimization silently strips epistemic marking. This is consistent with, and sharpens, verbosity- and sycophancy-style results [4, 5]: the pressure is not merely toward longer or more agreeable text but away from the specific surface forms by which pipelines currently signal uncertainty. Systems that must retain calibrated marking under preference pressure will need it carried somewhere preference cannot see — structured fields, tool outputs — rather than in-band prose.

Implication for evaluation. At auditable scale, five sixths of a raw preference verdict is reading order. Single-ordering LLM-judge evaluations at this scale are measuring position; debiased protocols are usable but must budget for a $\sim$0.35 decisive rate.

9 Limitations

One preference judge (which is also the writer; symmetric but family-specific), one aggregator family, 7–20B scale — the tension should be retested with larger judges and disjoint writer/judge families. The case battery is advisory-domain and English-only. The phrase-swap cell's null licenses only effects $\geq$0.22; the preference cell's decisive n (44) gives a minimum detectable share of $\sim$0.67, which the observed 0.818 clears but subtler lifts would not. Several early program values remain PROVISIONAL pending re-measurement and are not used here. Claims marked single-judge or single-writer should not be generalized past their scope.

10 Reproducibility

The measurement kit (`gst`: record contracts, pure-stdlib statistics, instruments, registry and gates), the frozen dictation registry and form-freeze files, all pre-registrations with their falsification conditions and attainability computations, every deviation and verdict, and the three program audits are in the project repository; each cell's registration is committed before its first run, so temporal ordering is verifiable from history.

References

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